DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
PRE BOARD EXAMINATION

CLASS: X | **FULL MARKS: 80**
SUBJECT: PHYSICS (SET A) | **TIME: 2 HOURS**

Candidates are allowed additional 15 minutes for only reading the paper.
They must NOT start writing during this time.
Section A is compulsory. Attempt any four questions from Section B
This paper consists of seven printed pages

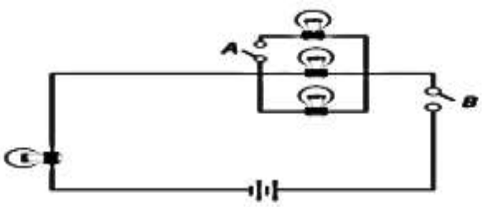
SECTION A

(Attempt all questions from this section)

1. Choose the correct answers to the questions from the given options: [15 × 1 = 15]

(i) Choose the correct statement. Latent heat absorbed is
- **(a) independent of the mass of the substance.**
- **(b) directly proportional to the specific heat capacity of the substance.**
- **(c) directly proportional to the increase in the temperature of the substance.**
- **(d) directly proportional to the specific latent heat of the substance.**

(ii) Study the circuit diagram. How many bulbs will light up when switch A is opened and switch B is closed?



(a) All 4 bulbs **(b) 3 bulbs** **(c) 1 bulb** **(d) 2 bulbs**

(iii) A wire of length 'L', made of material resistivity 'ρ' is cut into two equal parts. The resistivity of each of the two parts is equal to:
(a) ρ **(b) $\rho/2$** **(c) 2ρ** **(d) ρ^2**

(iv) The visible colour having the highest critical angle is
(a) red **(b) green** **(c) violet** **(d) yellow**

(v) An object is falling from a height of 10 m. When it falls by 7 m, which of the given statements is correct?
- **(a) Potential energy is greater than kinetic energy**
- **(b) Kinetic energy is greater than potential energy**
- **(b) Both types of mechanical energies are same**
- **(d) Total mechanical energy increases**

(vi) Two masses 1 g and 4 g are moving with equal kinetic energies. The ratio of the magnitudes of their linear momenta is
(a) 4:1 **(b) 1:2** **(c) 1:1** **(d) 1:6**

(vii) If a swimmer inside water looks at an aeroplane in the sky, then which of the following is correct?

(a) For the swimmer the aeroplane will appear to be lower than it actually is.
(b) For the swimmer the aeroplane will appear to be higher than it actually is.
(c) For the swimmer the aeroplane will appear at its actual height.
(d) For the pilot, the swimmer will appear to be at a greater height than he actually is.

(viii) A radioactive nucleus X decays with the emission of α particle as shown in the equation below.

$$ {}^{a}_{b}X \longrightarrow {}^{c}_{d}Y + \alpha $$

Choose the correct equation from the options given below:

(a) a = c (b) b = d + 1 (c) a = c + 4 (d) d = b + 1

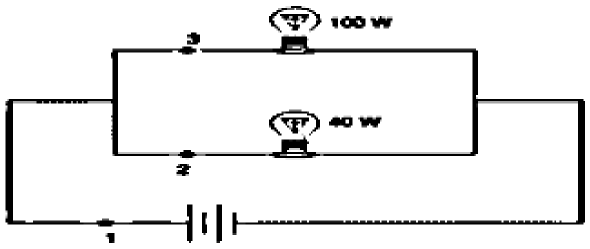
(ix) A convex lens is dipped in a liquid whose refractive index is equal to the refractive index of the lens. Then its focal length will

(a) become zero (b) become infinite (c) reduce (d) remain unchanged

(x) A wire pf length 80 cm has a frequency of 256 Hz. The length of a similar wire under similar tension, having frequency 1024 Hz is

(a) 320 cm (b) 20 cm (c) 40 cm (d) 160 cm

(xi) Choose the correct order of comparison of current flowing through points 1, 2 and 3.



(a) $I_1>I_2>I_3$ (b) $I_1>I_2< I_3$ (c) $I_1<I_2=I_3$ (d) $I_1>I_2=I_3$

(xii) Both UV rays and X-rays are subjected to strong electric field. Select the correct option.

(a) UV rays will be deflected more than X rays.
(b) UV rays will get less deflected compared to X rays.
(c) Both the radiation will get deflected to the same extent.
(d) None of the radiations will get deflected.

(xiii) With respect to uniform circular motion, select the correct option

(a) Acceleration remains constant.
(b) Velocity remains constant.
(c) No force acts on the object performing uniform circular motion.
(d) Work done by the force is zero during any interval.

(xiv) A wire in air stretched between two fixed supports is plucked exactly in the middle and then released. It executes

(a) resonant vibrations (b) natural vibration
(c) forced vibration (d) damped vibration

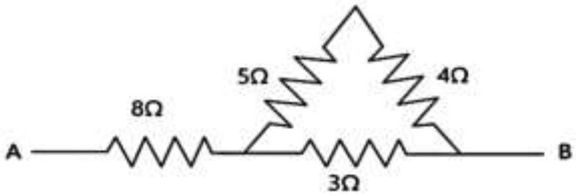
(xv) **Assertion (A) - A convex lens can be used as diverging lens.**
Reason (R) - The lens is placed in a medium with a higher index of refraction than that of the lens, hence the focal length becomes negative.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, and R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Question 2

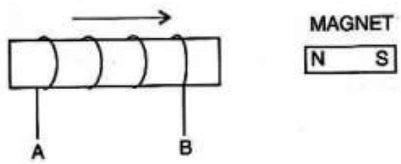
(i) (a) What do you mean by the statement "The heat capacity of a body is 60 J/K"?
(b) Name one factor on which the rise in temperature of a fuse wire depends.
(c) To which electrically charged plate the beta radiations will deflect while passing through an electric field? **[3]**

(ii) The specific heat capacity of a substance A is 3,800 $Jkg^{-1}\ K^{-1}$ and that of a substance B is 400 $Jkg^{-1}K^{-1}$. Which of the two substances is a good conductor of heat? Give a reason for your answer. **[2]**

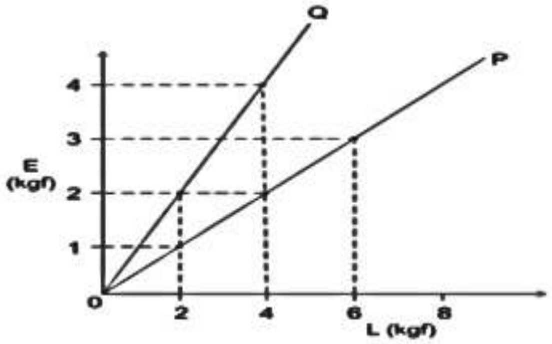
(iii) Calculate the effective resistance across AB. **[2]**



(iv) (a) State in which direction the current flows, A to B or B to A in the following diagram?
(b) How would the current in coil be altered, if the magnet was made to move with the same velocity as that of the coil? **[2]**



(v) With reference to the following diagram, identify the line (P or Q) that represents class II lever. Give reason in support of your answer **[2]**



(vi) A boy uses blue colour of light to find the focal length of a lens. He then repeats the experiment using red colour of light. Will the focal length be same or different in the two cases? Give a reason to support your answer. **[2]**

(vii) 50 g of metal piece at 27^0C requires 2400 J of heat energy so as to attain a temperature of 327^0C. Calculate the specific heat capacity of the metal. **[2]**

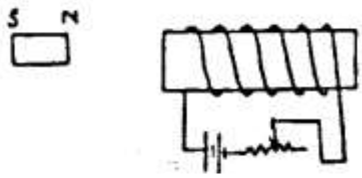
Question 3

(i) (a) Why is a nuclear fusion reaction called a thermo nuclear reaction?

(b) Complete the reaction: $^{3}He_{2} + {}^{2}H_{1} = {}^{4}He_{2}$ + ……………. + Energy **[2]**

(ii) Two metallic blocks P and Q having masses in the ratio 2:1 are supplied with the same amount of heat. If their temperatures rise by same degree, compare their specific heat capacity.

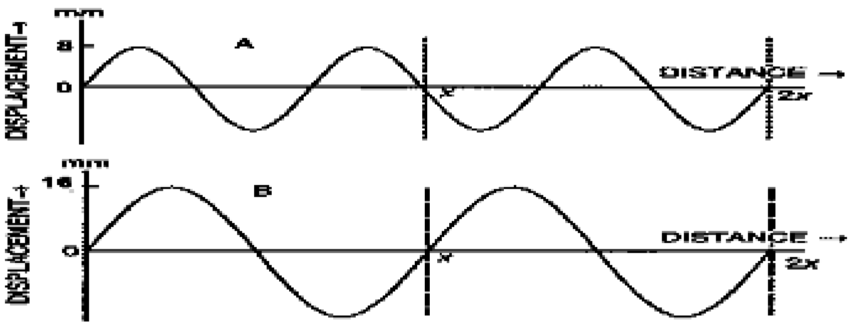
(iii) (a) With reference to the diagram given below, will the magnet get attracted or repelled?



(b) Mention any one way to increase the force mentioned in part (a). **[2]**

(iv) The displacement-distance graphs for waves A and B are depicted below as they pass through air. What are the ratios between their:

(a) Wavelengths and (b) Loudness? **[2]**



(v) A sound wave travelling in water has wavelength 0.5 m. Is this wave audible in air? Justify your answer. (The speed of sound in water = 1400 ms^{-1}) **[2]**

SECTION B

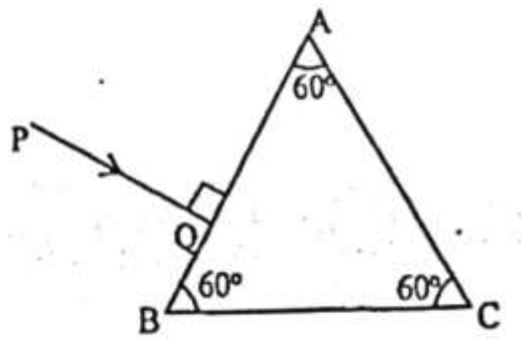
(Attempt any four questions)

Question 4

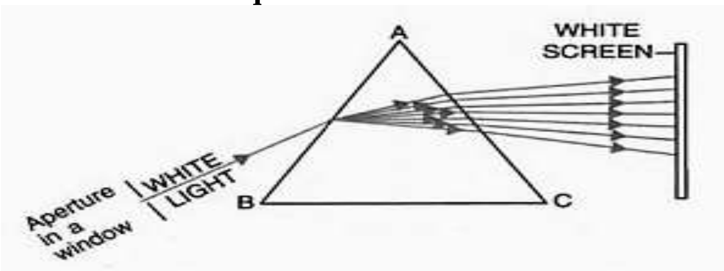
(i) a) Why does a light ray pass undeviated through the optical centre of a lens?

b) What are the reasons for preferring a total internal reflecting prism over a plane mirror in a periscope? **[3]**

(ii) Complete the path of the ray AB through the glass prism in PQR till it emerges out of the prism. Given the critical angle of the glass as 42°. Mark the angles as necessary. **[3]**



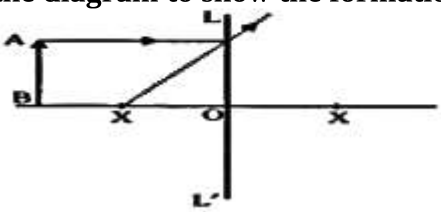
(iii) In the figure white light was allowed to pass through a triangular prism ABC. Study the figure the answer the questions that follow:



(a) Name the phenomenon occurring at the surface AB of the prism ABC.
(b) Define the phenomenon occurring at surface AC of the prism ABC.
(d) Which of the rays in the band formed on the white screen will suffer the least deviation?
(c) State the cause of the phenomenon mentioned by you in (a). [4]

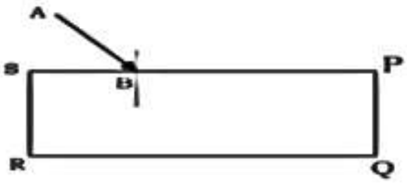
Question 5

(i) Copy and complete the diagram to show the formation of the image of the object AB.

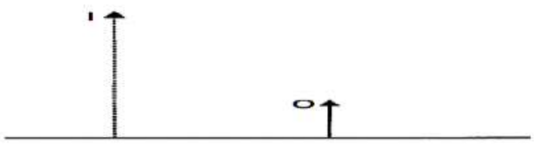


(b) What is the name given to X? [3]

(ii) (a) Redraw the diagram given below and complete the path of the ray AB till it emerges out of the rectangular glass block PQRS. In your diagram mark angle of incidence by the letter 'i' and angle of emergence 'e'.
(b) How are the angles 'i' and 'e' related? [3]



(iii) The given diagram shows an object O and its image I.



(a) Identify the lens.
(b) If the height of the image formed in the above diagram is 16 cm and that of the object kept at a distance 6 cm from the lens is 4 cm, find the position of the image and the focal length of the lens.

Question 6

(i) A simple pendulum, while oscillating, rises to a maximum vertical height of 20 cm from its rest position. If mass of the bob of the simple pendulum is 250 g, find
a) the velocity of the bob at its mean position given the pendulum is oscillating in friction less medium. (Take g = 10N/kg).
b) total mechanical energy during any instant of motion. [3]

(ii) **An uniform rule has its length 80 cm. It is supported on a knife edge placed at 10 cm mark. If the weight of the rule is 250 gf, what maximum weight can be placed at 0 cm mark? Draw the necessary diagram.** **[3]**

(iii) **A pulley system can lift a load of 1200 N by an effort of 250 N. If the resistance due to the weight of movable parts and friction is 300 N. Calculate**

a) mechanical advantage.

b) velocity ratio.

c) total number of pullies in the pulley system.

d) efficiency of the system. **[4]**

Question 7

(i) **A person is standing on the sea shore. An observer on the ship which in anchored in between a vertical cliff and the person on the shore fires a gun. The person on the shore hears two sounds, 2 s and 3 s after seeing the smoke of the fired gun. If the speed of the sound in the air is 320 ms^{-1}, then calculate**

(a) the distance between the observer on the ship and the person on the shore.

(b) the distance between the observer and the cliff. **[3]**

(ii) **A radioactive nucleus X emits an alpha particle followed by two beta particles and forms nucleus Y.**

a) What is the general name of the elements X and Y?

b) If the atomic mass number of Y is 189 then what is the mass number of X?

c) If the atomic number of Y is 80 then what is the atomic number of X? **[3]**

(iii) **Simon's father purchased a wooden vintage radio from an online shopping portal. Simon found two knobs on the radio, which are labelled as "Volume" and "Tuning" respectively.**

He wanted to explore the function of the tuning knob and rotated the same. He could hear different programmes being broadcast at different positions of the knob. He also noticed that the sound level of different programmes becomes maximum at some specific positions of the knob and that it gradually turns faint as the knob is rotated on either side.

He was curious and asked his elder brother, who was studying in Class X, about the possible reason for his observations.

a) Which phenomenon is responsible for hearing the loudest sound of a given programme at a specific position of the knob?

b) Define the phenomenon mentioned by you in a).

c) Give one more real-life example of the same phenomenon. **[4]**

Question 8

(i) **a) Why shouldn't we connect a fuse to the neutral wire?**

b) Why does the fuse provided with an electric pump has a current rating higher than the maximum current drawn by the pump?

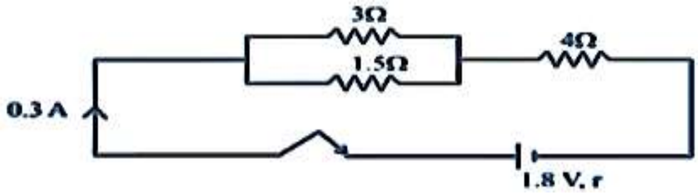
c) Why can't copper be used as a material to make fuse wires? **[3]**

(ii) **a) Rewrite and complete the following nuclear reaction by filling in the atomic number of Ba and mass number of Kr. Necessary mathematical calculations should be shown.**

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{144}_{\dots\dots}Ba + ^{\dots\dots}_{36}Kr + 3\,^{1}_{0}n + Energy$$

b) State one safety precaution in the disposal of nuclear waste. **[3]**

(iii) The diagram below shows three resistors connected across a cell of e.m.f. 1.8 V and internal resistance r.



Calculate:

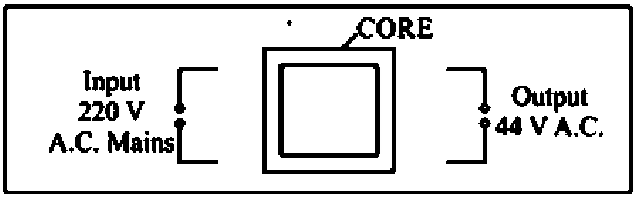
(a) current through 3 Ω resistor.

(b) the internal resistance r.

(c) potential difference across the terminal of the cell. [4]

Question 9

(i) The diagram below shows the core of a transformer and its input and output connections.



(a) Copy and complete the diagram of the transformer by drawing input and output coils.

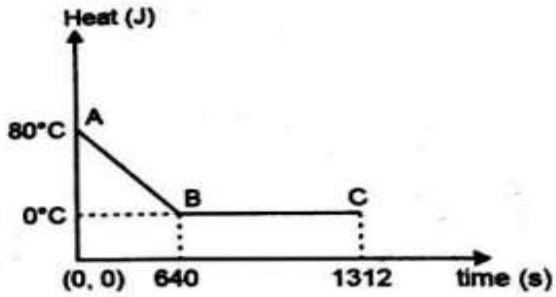
(b) What will be the frequency of the output voltage if the input voltage is applied at 50 Hz?

(c) Mention any one use of such transformers.

(ii) The diagram shows the cooling curve for 200 g of water. The heat is extracted at the rate of 100 J/s from the water. Answer the questions that follow:

(a) Calculate the specific heat capacity of water.

(b) Calculate the heat released in the region BC.



(iii) 104 g of water at 30°C is taken in a calorimeter made of copper of mass 42 g. When a certain mass of ice at 0°C is added to it, the final steady temperature of the mixture after the ice has melted, was found to be 10°C. Find the mass of ice added.

[Specific heat capacity of water = 4.2 $Jg^{-1}°C^{-1}$;

Specific latent heat of fusion of ice = 336 Jg^{-1};

Specific heat capacity of copper = 0.4 $Jg^{-1}°C^{-1}$] [4]